HVAC monitor — how the readings work

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I wanted to see whether I could get real sensor measurements off an HVAC system and onto a screen without having to be standing next to the unit. Right now I have the temperature part working. The current sensor is the next part I'm trying to figure out.

What I have working

A DS18B20 temperature probe is connected to my ESP32. The data wire goes to GPIO4. The ESP32 connects to my computer by USB, and the browser reads the incoming serial data. The live hardware page shows the temperature and graphs the readings as they come in.

Part	What it does in my setup
DS18B20	Measures temperature; powered by 3.3 V, data to ESP32 GPIO4, ground to GND.
ESP32	Reads the probe and sends the readings to my computer over USB.
Browser / website	Receives local USB serial frames, shows the latest temperature, and plots a live history.

What the data does

Temperature probe → ESP32 (GPIO4) → USB cable → browser serial connection → live temperature display and graph.

In the screenshot I saved, the site showed 75.76 °F / 24.31 °C. Another screen showed 143 validated frames in its local buffer and 0 rejected serial lines, with the graph changing as the probe temperature changed. Those are actual readings, not the example data used elsewhere in the demo website.

A problem I ran into

At one point I was getting 0.00 V when I checked a wire with a multimeter. I traced it to a connector that hadn't been crimped onto the wire properly. After fixing the connection, the meter read about 3.08 V. That taught me to check the physical connection before assuming the code or sensor was the problem.

What I check when something stops working

If I see…	I check…
No temperature value	USB connection, ESP32 power, probe wiring, GPIO4, then the serial output.
Zero voltage at a connection	Meter probes and settings, then the crimp and continuity of that wire.
Readings in serial but not on the page	Whether browser serial capture is running and whether the page is receiving valid frames.
Graph stops updating	Whether new USB frames arrive and the browser still has access to the device.

What I’m trying next

The second sensor is an SCT013 current clamp connected through an ADS1115 converter. I wired the ADS1115 to the ESP32 using SDA on GPIO21 and SCL on GPIO22. I have not gotten reliable current measurements yet, so I’m not using that data for any diagnosis.

My plan for diagnosing an HVAC problem

I’m not at the stage where the system can tell someone that a compressor, motor, or other part is failing. For now it measures and displays temperature. Once the current sensor works, I want to start comparing temperature changes with whether the equipment is actually drawing current.

My basic plan is to check whether the sensor is sending usable data first. Then I can look at how temperature changes over time. Eventually, with current readings and enough testing, I could check whether a unit is drawing power but not changing temperature the way I would expect. That would be a reason to investigate, not proof that one particular part has failed.

Status of each part

Working now	Still working on
ESP32 reads the DS18B20	Getting reliable SCT013 measurements through the ADS1115
USB sends readings to the computer	Testing the current circuit and its code
Website displays live temperature and a history graph	Developing and testing any diagnostic thresholds or alerts

Evidence I collected

01: Video: A separate demonstration of the build
02–03: Photos of my actual ESP32, breadboard, and wiring.
04: Photos showing the bad crimp, the zero-voltage reading, and the voltage after fixing it.
05: Wiring diagram with the temperature sensor connection and the unfinished current sensor marked clearly.
06: Diagram of the temperature data path through USB to the website.
07–08: Screenshots showing the live temperature graph and dashboard.

The live readings are received in the browser on the computer connected by USB. I have not shown that the hardware sends these readings directly to a remote server or over Wi-Fi.